

Chapter 8: Integration

A Level Pure Maths Notes · Rexponential

Why this chapter matters

Integration is differentiation run backwards: given a derivative, find the function it came from. This single idea underlies everything in this chapter, from finding a basic antiderivative to calculating the area under a curve, and it's why fluency in Chapter 7 pays off so directly here — every integration technique is really “which differentiation rule, run in reverse, gets me from this expression back to its antiderivative?” Integration of polynomials and basic definite integrals will feel familiar; substitution, integration by parts, and partial-fraction integration are the genuinely new material.

One conceptual anchor worth keeping throughout: a definite integral $\int_a^b f(x) dx$ is a *number* (typically representing an area), while an indefinite integral $\int f(x) dx$ is a *function* (plus an arbitrary constant, since many functions share the same derivative). Keeping this distinction clear prevents a lot of confusion about when a “+C” is needed.

8.1 Integration as the reverse of differentiation

Concept

Since differentiating x^n gives nx^{n-1} , integrating reverses this: to “undo” a derivative, raise the power by 1 and divide by the new power. Because differentiating any constant gives zero, integrating loses information about what constant was originally there — this is why every indefinite integral needs an arbitrary constant **+C**, representing the whole family of functions with that same derivative.

Procedure

For $n \neq -1$:

KEY FORMULA

- $\int x^n dx = x^{n+1}/(n+1) + C \quad (n \neq -1)$

Integrate a polynomial term by term, applying this rule to each term (including negative and fractional powers, once rewritten as x^n). A constant term k integrates to kx (since $k = kx^0$).

Worked example

EXAMPLE 8.1 (2 MARKS)

Find $\int (3x^2 - 4x + 2) dx$.

Step 1 Integrate term by term (M1): $\int 3x^2 dx = 3 \cdot (x^3/3) = x^3$; $\int -4x dx = -4 \cdot (x^2/2) = -2x^2$; $\int 2 dx = 2x$.

Answer $\int (3x^2 - 4x + 2) dx = x^3 - 2x^2 + 2x + C$ (A1). (Check: differentiating $x^3 - 2x^2 + 2x + C$ gives $3x^2 - 4x + 2$ — back to the original, confirming the integration.)

EXAM TIP

Forgetting the “+C” on an indefinite integral — this is an easy, cheap mark that's dropped surprisingly often; make writing +C an automatic habit every single time you integrate without limits.

8.2 Definite integrals and the area under a curve

Concept

A definite integral $\int_a^b f(x) dx$ calculates the (signed) area between the curve $y=f(x)$ and the x-axis, between $x=a$ and $x=b$. “Signed” matters: area where the curve is above the x-axis counts positively, area where it's below counts negatively — so a definite integral over a region straddling the x-axis does NOT give the total enclosed area unless you account for this (split the integral at the x-intercept and take the absolute value of each piece).

Procedure

The Fundamental Theorem of Calculus tells us:

KEY FORMULA

- $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$, where F is any antiderivative of f

In practice: find the indefinite integral (no +C needed here, since it cancels in the subtraction), substitute the upper limit, substitute the lower limit, and subtract.

Worked example

EXAMPLE 8.2 (4 MARKS)

Find the area under the curve $y = x^2$ between $x = 1$ and $x = 3$.

Step 1 $\int_1^3 x^2 dx = [x^3/3]_1^3$ (M1, finding the antiderivative).

Step 2 $= (3^3/3) - (1^3/3) = 27/3 - 1/3 = 26/3$ (M1, A1).

Answer Since $y=x^2 \geq 0$ throughout $[1,3]$, this is genuinely the area: **26/3 square units** (A1, stating the units and confirming the region is entirely above the axis).

EXAM TIP

Applying the definite integral formula to a region that dips below the x-axis without splitting it first — the negative and positive parts will partially cancel, giving a numerically smaller (and wrong) answer for the *area*, even though the integral's value itself is computed correctly.

8.3 The area between two curves

Concept

To find the area of the region trapped between two curves, integrate the *difference* between the upper curve and the lower curve, over the interval where that difference is non-negative.

Conceptually, this is the same as slicing the region into infinitely many thin vertical strips of height (top – bottom) and width dx , then summing them — exactly what an integral does.

Procedure

1. Find the x -values where the curves intersect (set the two expressions for y equal and solve) — these become your limits of integration.
2. Determine which curve is on top over that interval (check with a test value, or from the shapes of the graphs).
3. Integrate (top – bottom) between the intersection points: $\text{Area} = \int_a^b [f(x) - g(x)] dx$, where f is the upper curve.

Worked example

EXAMPLE 8.3 (6 MARKS)

Find the area of the region enclosed between the curves $y = x^2$ and $y = x + 2$.

Step 1 Find intersections: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0 \Rightarrow x = -1$ or $x = 2$ (M1, A1).

Step 2 Test $x=0$ (between the intersections): $y = x+2$ gives 2, $y = x^2$ gives 0, so the line $y=x+2$ is on top over $[-1,2]$ (M1).

Step 3 $\text{Area} = \int_{-1}^2 [(x+2) - x^2] dx = [x^2/2 + 2x - x^3/3]_{-1}^2$ (M1).

Step 4 At $x=2$: $4/2 + 4 - 8/3 = 2+4-8/3 = 6 - 8/3 = 10/3$. At $x=-1$: $1/2 - 2 - (-1/3) = 0.5 - 2 + 1/3 = -1.5 + 1/3 = -7/6$ (A1, both evaluations correct).

Answer $\text{Area} = 10/3 - (-7/6) = 20/6 + 7/6 = 27/6 = 9/2 = 4.5$ square units (A1).

EXAM TIP

Integrating (bottom – top) instead of (top – bottom), which gives a negative value — area must be positive, so if your calculation comes out negative, you've either mismatched the order or misidentified which curve is on top; check with a test point rather than guessing.

8.4 The trapezium rule

Concept

Not every function can be integrated using a neat algebraic antiderivative (many genuinely have none expressible in elementary functions). The trapezium rule is a **numerical** method for estimating a definite integral: it approximates the area under a curve by dividing it into a series of thin vertical strips and treating the top of each strip as a straight line (a trapezium), rather than the true curve. More strips (a smaller width h) gives a more accurate approximation.

Procedure

To estimate $\int_a^b f(x) dx$ using n equal strips of width $h = (b-a)/n$:

KEY FORMULA

- $\int_a^b f(x) dx \approx (h/2)[y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})]$

where $y_0, y_1, \dots, y_n$ are the function values at the $n+1$ equally-spaced x -values from a to b (the endpoints y_0 and y_n are counted once, all the interior points are counted with a coefficient of 2).

Worked example

EXAMPLE 8.4 (4 MARKS)

Use the trapezium rule with 4 strips to estimate $\int_0^4 \sqrt{1+x^3} \, dx$, giving your answer to 3 significant figures.

Step 1 With 4 strips over $[0,4]$, $h = 4/4 = 1$, so $x = 0, 1, 2, 3, 4$ (M1).

Step 2 Compute y -values: $y_0=\sqrt{1}=1$, $y_1=\sqrt{2}=1.4142$, $y_2=\sqrt{9}=3$, $y_3=\sqrt{28}=5.2915$, $y_4=\sqrt{65}=8.0623$ (M1, A1 for correct values to reasonable accuracy).

Answer Apply the rule: $\int \approx 1/2 \times [y_0 + y_4 + 2(y_1+y_2+y_3)] = \frac{1}{2} \times [1 + 8.0623 + 2(1.4142+3+5.2915)] = \frac{1}{2} \times [9.0623 + 2(9.7057)] = \frac{1}{2} \times [9.0623 + 19.4114] = \frac{1}{2} \times 28.4737 = \mathbf{14.2}$ (3 s.f.) (A1).

EXAM TIP

Forgetting to double the interior y -values (or accidentally doubling the endpoints too) — write out the coefficient pattern explicitly (1, 2, 2, ..., 2, 1) before substituting numbers, rather than trying to apply the formula from memory under time pressure.

8.5 Integrating exponentials, $1/x$, and trig functions

Concept

These are the direct reverses of the derivatives you learned in 7.5 and 7.6 — for every differentiation rule, there's a matching integration rule running the other way. Since e^x differentiates to itself, it also integrates to (almost) itself; since $1/x$ came from differentiating $\ln x$, integrating $1/x$ gives $\ln x$ back (with an absolute value, since $\ln$ is only defined for positive arguments but $1/x$ is defined for negative x too).

Procedure

KEY FORMULAS

- $\int e^{kx} \, dx = (1/k)e^{kx} + C$
- $\int 1/x \, dx = \ln|x| + C$
- $\int \sin(kx) \, dx = -(1/k)\cos(kx) + C$
- $\int \cos(kx) \, dx = (1/k)\sin(kx) + C$

Notice the pattern: integrating introduces a $1/k$ factor wherever differentiating introduced a k — they're inverse operations, so this is exactly what you'd expect.

Worked example

EXAMPLE 8.5 (5 MARKS)

Find (a) $\int e^{2x} dx$, (b) $\int (1/x) dx$, (c) $\int \sin(3x) dx$.

Step 1 (a) $\int e^{2x} dx = \frac{1}{2} e^{2x} + C$ (M1, A1 — check: differentiating $\frac{1}{2}e^{2x}$ gives $\frac{1}{2} \times 2e^{2x} = e^{2x}$ ✓).

Step 2 (b) $\int (1/x) dx = \ln|x| + C$ (B1).

Answer (c) $\int \sin(3x) dx = -\frac{1}{3}\cos(3x) + C$ (M1, A1 — check: differentiating $-\frac{1}{3}\cos(3x)$ gives $-\frac{1}{3} \times (-3\sin 3x) = \sin 3x$ ✓).

EXAM TIP

Forgetting the $1/k$ scaling factor — writing $\int e^{2x} dx = e^{2x} + C$ (missing the $\frac{1}{2}$), a very natural slip since it's easy to think “integration undoes differentiation” without remembering that the chain-rule constant needs to be divided out, not just left alone.

8.6 The reverse chain rule

Concept

The chain rule (7.7) says: differentiating a composite function multiplies the outer derivative by the inner derivative. Run in reverse, this means: whenever you spot an expression of the form (something)ⁿ multiplied by the derivative of that “something,” you can integrate it directly by reversing the power rule on the outer part — no substitution machinery needed once you recognise the pattern. A special case of this, $\int f'(x)/f(x) dx = \ln|f(x)| + C$, comes from recognising $f(x)^{-1} \times f'(x)$ as this same pattern with $n = -1$.

Procedure

KEY FORMULAS

- $\int [g(x)]^n \times g'(x) dx = [g(x)]^{(n+1)} / (n+1) + C$, for $n \neq -1$
- $\int f'(x)/f(x) dx = \ln|f(x)| + C$

The key recognition skill: look for a function and, elsewhere in the same expression, something that is (a constant multiple of) its derivative.

Worked example

EXAMPLE 8.6 (4 MARKS)

Find (a) $\int 2x(x^2+1)^4 dx$, (b) $\int 2x/(x^2+1) dx$.

Step 1 (a) Let $g(x) = x^2+1$, so $g'(x) = 2x$ — which is exactly the other factor present (M1, spotting the pattern). Using $\int [g(x)]^4 g'(x) dx = [g(x)]^5/5 + C$: $\int 2x(x^2+1)^4 dx = (x^2+1)^5/5 + C$ (A1).

Answer (b) Let $f(x) = x^2+1$, so $f'(x) = 2x$ — again exactly the numerator (M1). Using $\int f'(x)/f(x) dx = \ln|f(x)| + C$: $\int 2x/(x^2+1) dx = \ln|x^2+1| + C$, or simply $\ln(x^2+1) + C$ since $x^2+1 > 0$ always (A1).

EXAM TIP

Trying to apply this shortcut when the “extra factor” is only a partial or incorrect match for the derivative — e.g. attempting it on $\int x(x^2+1)^4 dx$ (missing the factor of 2) without adjusting for the missing constant; if the match isn't exact, you can often fix it by pulling out a compensating constant (e.g. writing $x(x^2+1)^4 = \frac{1}{2} \times 2x(x^2+1)^4$), but always check this adjustment carefully rather than assuming the shortcut applies unchanged.

8.7 Integration by substitution

Concept

Integration by substitution is the general-purpose version of the reverse chain rule from 8.6, used when the pattern isn't quite so immediately obvious. The idea: replace part of the expression with a new variable u , chosen so that after substitution the whole integral is written purely in terms of u (with dx also converted to du) — turning a hard integral in x into an easier one in u .

Procedure

1. Choose a substitution $u = g(x)$ — usually the “inner” function, or whatever's making the integral awkward.
2. Find du/dx , then rearrange to express dx in terms of du (e.g. $du = g'(x) dx$, so $dx = du/g'(x)$).
3. Rewrite the entire integral in terms of u only (every x , and dx , must disappear).
4. Integrate with respect to u , then substitute $u = g(x)$ back at the end to return to a function of x .
5. For a **definite** integral, you can either substitute back and use the original x -limits, or (often faster) convert the limits themselves into u -values at step 2 and never substitute back at all.

Worked example

EXAMPLE 8.7 (4 MARKS)

Find $\int x\sqrt{x^2+1} dx$ using the substitution $u = x^2+1$.

Step 1 $u = x^2+1 \implies du/dx = 2x \implies du = 2x dx \implies x dx = \frac{1}{2} du$ (M1, rearranging to isolate $x dx$, which is what appears in the integral).

Step 2 Rewrite the integral: $\int x\sqrt{x^2+1} dx = \int \sqrt{u} \cdot (\frac{1}{2} du) = \frac{1}{2} \int u^{1/2} du$ (M1, substitution complete — no x remains).

Step 3 Integrate: $\frac{1}{2} \times [u^{3/2}/(3/2)] + C = \frac{1}{2} \times (2/3)u^{3/2} + C = \frac{1}{3} u^{3/2} + C$ (M1).

Answer Substitute back $u = x^2+1$: $\frac{1}{3}(x^2+1)^{3/2} + C$ (A1).

EXAM TIP

Leaving a stray x in the integral after substitution — e.g. forgetting to convert $x dx$ fully into an expression in u alone; if any x survives (other than inside an expression you're deliberately about to convert), the substitution isn't finished, and you cannot integrate with respect to u until every trace of x is gone.

8.8 Integration by parts

Concept

Integration by parts is the reverse of the product rule (7.8), used for integrating a **product** of two different types of function (like x and e^x , or x and $\sin x$) where substitution doesn't obviously apply. It doesn't fully "solve" the integral in one step — instead, it trades the original integral for a hopefully-easier one.

Procedure

The formula (derived from the product rule):

KEY FORMULA

- $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$

(often abbreviated $\int u dv = uv - \int v du$). The skill is choosing which factor is u and which is dv/dx : as a rule of thumb, choose u to be the factor that gets **simpler** when differentiated (like a polynomial, which eventually differentiates to a constant), and dv/dx to be the factor that's straightforward to integrate (like e^x or $\sin x$, which don't get more complicated).

Worked example

EXAMPLE 8.8 (4 MARKS)

Find $\int x e^x dx$.

Step 1 Choose $u = x$ (differentiates to a constant, getting simpler) and $dv/dx = e^x$ (easy to integrate) (M1, justifying the choice).

Step 2 Then $du/dx = 1$, and $v = e^x$ (M1, integrating dv/dx ; no $+C$ needed at this intermediate step).

Step 3 Apply the formula: $\int x e^x dx = x \cdot e^x - \int e^x \cdot 1 dx = x e^x - \int e^x dx$ (M1).

Answer $= x e^x - e^x + C = e^x(x - 1) + C$ (A1, factorised).

EXAM TIP

Choosing u and dv/dx the wrong way round — e.g. picking $u = e^x$ and $dv/dx = x$ here would leave you needing to integrate $x \cdot e^x/2 dx$ by parts again, going in circles rather than making progress. If your first attempt makes the resulting integral *harder* rather than easier, that's the signal to swap your choice of u and dv/dx .

8.9 Integration using partial fractions

Concept

Some rational functions (fractions of polynomials) can't be integrated directly using any of the previous techniques, but can be integrated easily once they're split into **partial fractions** (2.6) — simpler fractions, each of which integrates using the standard $\int 1/x dx = \ln|x| + C$ pattern from 8.5. This is a good example of how a Chapter 2 algebra technique becomes directly useful for calculus later on.

Procedure

1. Split the given rational function into partial fractions, exactly as in section 2.6.
2. Integrate each partial fraction separately — each one will typically be of the form $k/(x+a)$, which integrates to $k \cdot \ln|x+a| + C$.
3. Combine the results (and the constants) into a single expression, with one overall $+C$.

Worked example

EXAMPLE 8.9 (6 MARKS)

Find $\int 1/[(x+1)(x+2)] dx$.

Step 1 Split into partial fractions: $1/[(x+1)(x+2)] = A/(x+1) + B/(x+2)$. Multiplying through: $1 = A(x+2) + B(x+1)$ (M1).

Step 2 Let $x = -1$: $1 = A(1) \implies A = 1$. Let $x = -2$: $1 = B(-1) \implies B = -1$ (M1, A1 for both constants).

Step 3 So $1/[(x+1)(x+2)] = 1/(x+1) - 1/(x+2)$ (A1).

Answer Integrating each term: $\int [1/(x+1) - 1/(x+2)] dx = \ln|x+1| - \ln|x+2| + C$, which can also be written as $\ln|(x+1)/(x+2)| + C$ using the laws of logs (M1, A1).

EXAM TIP

Attempting to integrate the original unsplit fraction directly using $\int 1/x dx = \ln|x| + C$ — that formula only applies to a *linear* denominator like $(x+a)$, not a product of two linear factors; partial fractions must be found first.

8.10 Forming and solving differential equations

Concept

A differential equation is an equation involving a derivative (like dy/dx) — rather than being told y directly as a function of x , you're told how y *changes*, and asked to reconstruct y itself. Many real situations are naturally described this way (rate of population growth depends on current population, rate of cooling depends on current temperature difference, and so on). **Separation of variables** is a technique for solving the particular type of differential equation where dy/dx can be rearranged into “a function of y ” times “a function of x ” separately.

Procedure

For a differential equation of the form $dy/dx = f(x)g(y)$:

1. Separate the variables: rearrange so all the y -terms (including dy) are on one side, and all the x -terms (including dx) are on the other: $(1/g(y)) dy = f(x) dx$.
2. Integrate both sides independently: $\int (1/g(y)) dy = \int f(x) dx$.
3. This gives an equation relating y and x , generally with one overall arbitrary constant (the two constants from each side's integration combine into one).

4. If given an initial condition (a known point the solution passes through), substitute it in to find the specific value of the constant, giving the **particular solution** (as opposed to the general solution, which still has the arbitrary constant in it).

Worked example

EXAMPLE 8.10 (6 MARKS)

Solve the differential equation $dy/dx = xy$, given that $y = 2$ when $x = 0$. Give y as a function of x .

Step 1 Separate the variables: $(1/y) dy = x dx$ (M1, dividing both sides by y and multiplying by dx — valid since y is not identically zero here).

Step 2 Integrate both sides: $\int (1/y) dy = \int x dx \implies \ln|y| = x^2/2 + C$ (M1, A1, one constant is enough since we can absorb both sides' constants into one).

Step 3 Exponentiate both sides to solve for y : $|y| = e^{(x^2/2 + C)} = e^C \cdot e^{(x^2/2)}$. Writing $A = e^C$ (a new positive constant, or allowing y to be negative absorbs the modulus): $y = A \cdot e^{(x^2/2)}$ (M1).

Step 4 Apply the initial condition $y=2$ when $x=0$: $2 = A \cdot e^0 = A \cdot 1 \implies A = 2$ (M1).

Answer So the particular solution is **$y = 2e^{(x^2/2)}$** (A1).

EXAM TIP

Forgetting the constant of integration when first integrating (or adding two separate, unmerged constants on each side rather than one combined constant) — and, separately, forgetting to apply the initial condition at all, which leaves the *general* solution (still containing an unknown constant) when a *particular* numerical solution was asked for.

Chapter summary

Key formulas

POWER RULE

- $\int x^n dx = x^{(n+1)}/(n+1) + C \quad (n \neq -1)$

EXPONENTIAL AND 1/X

- $\int e^{kx} dx = (1/k)e^{kx} + C$
- $\int 1/x dx = \ln|x| + C$

TRIG

- $\int \sin(kx) dx = -(1/k)\cos(kx) + C$
- $\int \cos(kx) dx = (1/k)\sin(kx) + C$

REVERSE CHAIN RULE

- $\int [g(x)]^n g'(x) dx = [g(x)]^{(n+1)/(n+1)} + C$
- $\int f'(x)/f(x) dx = \ln|f(x)| + C$

INTEGRATION BY PARTS

- $\int u dv = uv - \int v du$

TRAPEZIUM RULE

- $\int_a^b f(x) dx \approx (h/2)[\text{first} + \text{last} + 2(\text{sum of the rest})]$

SEPARATION OF VARIABLES

- $dy/dx = f(x)g(y) \implies \int (1/g(y)) dy = \int f(x) dx$

Common exam mistakes across this chapter

- Forgetting the “+C” on indefinite integrals.
- Applying a definite integral formula to a region that dips below the x-axis without splitting it, when asked for an *area* rather than the integral's raw value.
- Forgetting the $1/k$ scaling factor when integrating e^{kx} , $\sin(kx)$, $\cos(kx)$.
- In integration by parts, choosing u and dv the wrong way round, making the resulting integral harder rather than easier.
- Trying to integrate a fraction with a factorised denominator directly, instead of splitting into partial fractions first.
- Forgetting to apply a given initial condition, leaving a general solution when a particular one was required.

You should now be able to

- Integrate polynomials, exponentials, $1/x$ and trig functions, remembering +C on every indefinite integral
- Find the area under a curve, the area between two curves, and estimate an integral with the trapezium rule
- Integrate using the reverse chain rule, substitution, integration by parts, and partial fractions
- Form and solve a differential equation by separating the variables, including finding a particular solution